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# Hierarchical medical image report adversarial generation with hybrid discriminator☆

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ABSTRACT

*Background and objectives:* Generating coherent reports from medical images is an important task for reducing doctors' workload. Unlike traditional image captioning tasks, the task of medical image report generation faces more challenges. Current models for generating reports from medical images often fail to characterize some abnormal findings, and some models generate reports with low quality. In this study, we propose a model to generate high-quality reports from medical images.

*Methods:* In this paper, we propose a model called Hybrid Discriminator Generative Adversarial Network (HDGAN), which combines Generative Adversarial Network (GAN) with Reinforcement Learning (RL). The HDGAN model consists of a generator, a one-sentence discriminator, and a one-word discriminator. Specifically, the RL reward signals are judged on the one-sentence discriminator and one-word discriminator separately. The one-sentence discriminator can better learn sentence-level structural information, while the one-word discriminator can learn word diversity information effectively.

*Results:* Our approach performs better on the IU-X-ray and COV-CTR datasets than the baseline models. For the ROUGE metric, our method outperforms the state-of-the-art model by 0.36 on the IU-X-ray, 0.06 on the MIMIC-CXR and 0.156 on the COV-CTR.

*Conclusions:* The compositional framework we proposed can generate more accurate medical image reports at different levels.

## 1. Introduction

Unlike the traditional image captioning task [1,2], the task of generating medical image report aims to generate topic-coherent and informative reports rather than brief descriptions composed of sentences. Recently, the medical image report generation task attracts increasing attention [3] from the computer vision community due to its wide range of applications and assistance to radiologists and pathologists. Specifically, reading medical images and writing medical image reports can be error-prone for inexperienced physicians, and time-consuming for experienced physicians in real life. As one of the most representative and practical report generation tasks, the task of generating medical image reports can bring physicians convenience and alleviate their workload. However, it is challenging for machines to extract the important meanings from medical image pixels and to generate correct and fluent reports. Particularly, the generated report must comply with more critical protocols. As shown in Fig. 1, a complete medical image report consists of impression, tags, and findings describing the detailed diagnostic results related to both normal and abnormal features. Therefore, we propose a Hybrid Discriminator Generative Adversarial Network (HDGAN), which is the first attempt to incorporate Generative Adversarial Network (GAN) to handle more

☆ The code of this paper: https://github.com/UpcPaper/HDGAN.

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**Medical Image Report**
**Impression:** no acute cardiopulmonary process.
**Findings:** coronary artery stents visualized. the cardiomediastinal silhouette is within normal limits for size and contour. the lungs are normally inflated without evidence of focal airspace disease pleural effusion or pneumothorax. no acute bone abnormality.
**Tags:** right atrium

**Fig. 1.** An exemplar chest X-ray report.

complicated tasks of medical image report generation and help physicians reduce their workload.

As shown in Fig. 1, a complete medical image report is composed of more text information, including impression, tags, and findings that comply with more critical protocols and involve the use of many medical terms. As a result, the image captioning models, which often output brief sentences, tend to perform poorly in executing such tasks. In a word, we cannot simply copy image captioning models to perform medical image report generation tasks directly. Intrinsically, the key and the most significant difficulty for handling a medical image report generation task is how to output a long and correct narrative report. On the other hand, to generate a medical image report, some medical knowledge is needed [4]: (1) the normal anatomy of the thorax and the basic physiology of chest diseases; (2) clinical presentation and history; and (3) the correlation with other diagnostic results (electrocardiogram and respiratory function tests). There are many works dedicated to medical image report generation tasks, and great progresses have been made in this area [5,6]. Some medical image generation models tend to perform poorly in executing medical image report generation tasks subject to specific content requirements due to several reasons. For instance, as shown in Fig. 2, the existing generative model tends to generate some repeated sentences and wrong results, especially wrong abnormal findings in some cases. However, in medical image reports, the abnormal findings are few but more important. It is difficult for medical image report generation models to comprehend the esoteric principles of medical knowledge.

First, a complete medical image report is a long narrative text description consisting of multiple sentences or paragraphs, so more attention needs to be given to text processing in the process of medical image report generation to further improve the performance of the algorithm. Second, in particular, medical image reports are composed of not only a large number of normal findings, but also significant abnormal findings. Therefore, the performance of a medical image report generation models needs to be enhanced so that it can correctly generate high-quality reports, especially the abnormal findings. For instance, as shown in Fig. 2, the existing medical image report generation models generate repeated sentences and wrong findings, because they only emphasize the encoding-decoding process without paying attention to the text processing.

In order to address the issues described above, a method for hierarchical adversarial generation of medical image reports with a hybrid discriminator, which is called Hybrid Discriminator Generative Adversarial Network (HDGAN), is proposed in this paper. We use GAN [7] and RL [8] to execute this task with more attention to text processing. It is well known that GANs have enjoyed considerable success and have been mostly used to execute computer vision tasks, such as generating samples of natural images [9]. Nonetheless, the discrete outputs make it difficult to transfer the gradient update from the discriminator to the generator when a GAN is used for generating sequences of discrete tokens such as text [10]. Therefore, we introduce RL to solve this problem. Furthermore, we use two kinds of rewards from different discriminators to obtain better information with different gradients for updating the generator. One discriminator is called one-sentence discriminator, which can learn the sentence structure information and use generated sentences as its input. The other decimator is called one-word discriminator, which can learn the word diversity information and use generated words as its input. The main contributions of this paper can be summarized below.

- A GAN is used to improve the quality of medical image reports. RL is incorporated into the process of the GAN. Specifically, the generator is regarded as an agent in reinforcement learning, the state is defined

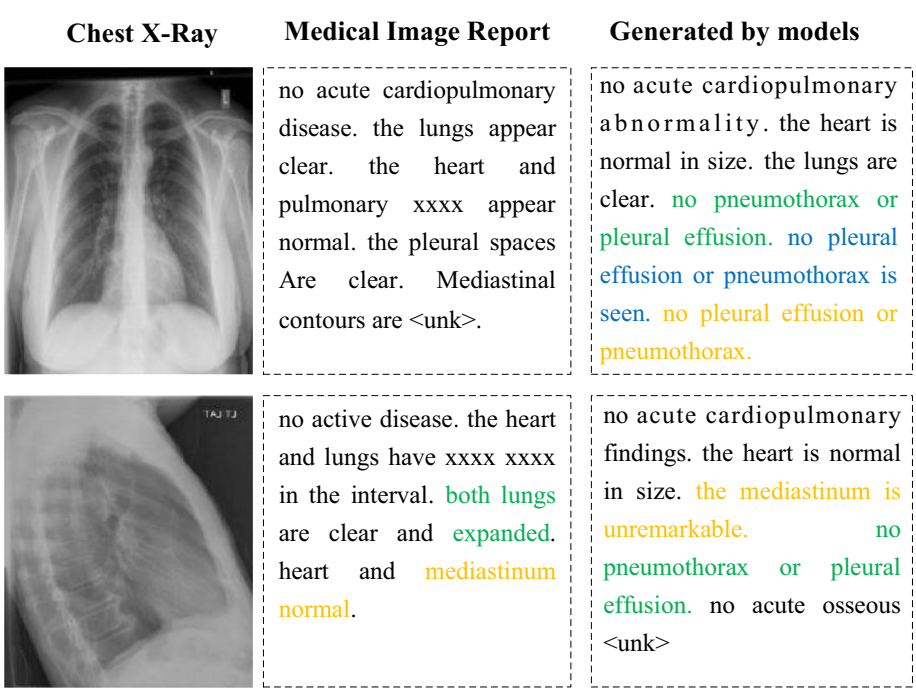



**Fig. 2.** Existing model generation results display: the upper part is the display of generated repeated sentences; the lower part is the display of generated wrong results.

as the current generated sequence, and the action is defined as the next generated character.

- Two kinds of discriminators are used to obtain the reward value. One discriminator is called one-sentence discriminator, whose reward can guide the generator to output more structured and logical medical image reports. The other discriminator is called one-word discriminator whose reward can guide the generator to output more diversified and special-purpose words.
- Comprehensive experiments were conducted to analyze the proposed method. The results on the IU-X-ray, MIMIC-CXR and COV-CTR dataset show that the proposed method delivers a competitive performance. The reports generated using this method meet the requirements for medical image reports, such as correct terminology and hierarchical structure.

The remaining sections of this paper are organized as follows. In Section 2, we provide some background knowledge related to image captioning task, medical image report generation task, GAN and RL used for generating text. In Section 3, we first give an overview of the proposed model and then present each module of the proposed model in detail. In Section 4, we describe the datasets used in our experiments and provide the experimental results and analysis results. Finally, we conclude this paper in Section 5.

## 2. Related work

### 2.1. Image captioning with deep learning

Image captioning is a comprehensive task that combines computer vision and natural language processing. It requires not only the use of models to understand the content of the images, but also the use of natural language to express the relationship between them. In addition, the model also needs to be capable of capturing the semantic information of the image and generate human-readable sentences. Most of the current image captioning methods are based on the encoder-decoder model [1]. The encoder is generally a convolutional neural network, which uses the features of the final fully connected layer or convolutional layer as the image feature. The decoder is generally a recurrent neural network, which is mainly used to generate image descriptions. Additionally, many attention-based deep neural networks have been proposed for image captioning. Typically, some models can be characterized as top-down approaches, and the top-down visual attention mechanisms [11] are able to learn deeper image understanding through fine-grained analysis and even multiple steps of reasoning [12,13]. A combined bottom-up and top-down attention mechanism used in image captioning task enables attention to be calculated at the level of objects and other salient image regions [14] so as to achieve better result. Additionally, in order to improve the diversity and naturalness of the generated captions, some works that introduce GAN [15,16] have been done. In these works, the discriminator uses the generated sentences as the input to judge whether they are real or fake. At present, the state-of-the-art approaches based on the CNN-RNN architectures and using ASG [17] or multi-task triple-stream networks [18] were proved to be effective and prominent in the domain of computer vision. Recent work showed that in terms of efficiency and performance, Transformer is superior to LSTM in some NLP tasks, so the work [19] used Transformer to substitute the traditional decoder LSTM to accelerate the training process and combined the spatial attention and adaptive attention into Transformer, which enabled the decoder to determine where and when to use image region information. Recent work on image captioning focused on generating multiple captions to describe an image. The work [20] used the word-to-word Hamming distance in the high-dimensional output space of captions to generate diverse captions. Furthermore, there are some works that leverage generative adversarial networks (GANs) [21] or Variational Autoencoders (VAEs) to mitigate the limitations of sampling in high-dimensional spaces. Since GANs for generating diverse captions has the problem of limited accuracy, recent works used VAEs [22–24] to learn latent representations conditioned on images to sample diverse captions. Similarly, the image processing and report generation process in medical images report generation tasks are similar to those of some methods used to execute image captioning tasks.

However, these works were performed based on natural scenarios and were limited by a lack of targeted learning of professional medical vocabulary. Therefore, we propose a one-word reward method to solve this problem. It encourages the model to learn medical vocabulary and generates more professional reports from medical images.

### 2.2. Medical images report generation

Compared with the outputs of image captioning tasks, medical report generation tasks have informative outputs with multiple, heterogeneous sentences. In other words, medical report generation tasks pose higher requirements for content selection, relation generation, and content ordering. There are several recent works involved in the generation of reports based on given medical images. Most of these works are based on the encoder-decoder architecture. The hybrid retrieval generation reinforced agent (HRGR-Agent) [5] harmonized traditional retrieval methods based on human prior knowledge with the methods based on RL to generate structured and diversified reports. The Knowledge-driven Encode, Retrieve, Paraphrase (KERP) [6] approach combined knowledge retrieval-based methods with learning-based methods to generate accurate, robust medical image reports. The text-image embedding network [25], which is a multi-level attention model, was integrated into an end-to-end, trainable CNN-RNN architecture to highlight meaningful text and image regions. The network was mainly used to perform the classification of common chest diseases, and it could also generate reports. The multimodal model [26] used a recurrent attention mechanism to combine the encoding of the image and one generated sentence to guide the generation of the next sentence. In the newest method [27], a "memory-driven Transformer [28]" was proposed to generate radiology reports, where a relational memory was designed to record key information in the generation process and a memory-driven conditional layer normalization was applied to incorporate the memory into the decoder of the Transformer. A reinforced medical report generation solution with x-linear attention and repetition penalty mechanisms (ReMRG-XR) [29] used the x-linear attention modules to explore high-order feature interactions and achieve multi-modal reasoning and applied penalties to repeated terms during the model's training process. CA [30] proposed a contrastive attention (CA) model to effectively capture and describe abnormal regions. NSL [31] interpolated a symbolic graph reasoning module into the network through embedding prior domain knowledge.

The works mentioned above have made great contributions to the advancement of automatic generation of reports from medical images. However, there is still a lack of comprehensive word-level and sentence-level guidance. Therefore, we propose a reward fusion method to provide comprehensive word-level and sentence-level guidance for the generation of medical image reports.

### 2.3. GAN and RL used for generating text

GANs have shown considerable success and have mostly been applied to computer vision tasks, such as generating samples of natural images of which data representation is continuous. When it comes to the discrete data representation, the GANs' drawback is exposed because the discrete outputs generated by the generator make it difficult to transfer the gradient update from the discriminator to the generator. So far, there have been some research results achieved in applying GAN to sequence discrete data generation. For example, for modeling the sequence generator as a policy of picking the next token, policy gradient methods [32] can be adopted to optimize the generator once there is an (implicit) reward function to guide the policy. SeqGAN [33] applied the idea of

GAN and RL, the reward generated by reinforcement learning was used to replace the gradient descent of continuous variables, and the rollout module of the generator was used to avoid the instability of the deep reinforcement learning network. TextGAN [34] used a CNN as the discriminator and used the idea of smooth approximation to approximate the output of the generator LSTM so as to solve the problem of non-differentiation of gradient caused by dispersion. The optimization function used the feature matching method and a variety of training methods to improve the convergence of the GAN. It is well known that mode collapse [35] often occurs in the training progress of GAN. The work [36] proposed to use the kernelized discrepancy metric to match the hidden feature representations of real sentences and generated sentences. This method can well mitigate the mode collapse problem in GANs' training progress. MASKGAN [37] used the actor-critic algorithm in reinforcement learning to train the generator and used the maximum likelihood and stochastic gradient descent to train the discriminator. The method of using the scene based on the context to fill the missing words avoided the problem of mode collapse and training instability, which are problems that a GAN is prone to, in text generation tasks.

The works mentioned above have improved the representational ability of GANs on discrete datasets. However, they still have two limitations. First, the methods of GANs are based on common scenarios, and there is a lack of special applications in specific fields, such as medical diagnosis. Second, the methods of medical report generation work on the same level to study the text representation and lack exploration at different levels. Inspired by these works, we design a Hybrid Discriminator Generative Adversarial Network to relate GANs to the field of medical diagnosis and learn the representation of medical image reports at different levels.

## 3. Our approach

### 3.1. Overview

Fig. 3 shows the whole structure of the HDGAN model. First, the fake data generated by the generator and the real data from the medical image dataset are input into the hybrid discriminator that includes a one-sentence discriminator and a one-word discriminator. Second, the two discriminators output the results as rewards, which are regarded as feedback for the generator. Specifically, the discriminator computes the facticity of the generated sequence and determines whether it satisfies the expectation of the generator. Higher probability values imply more beneficial reward fusion, and lower probability values imply more detrimental reward fusion. It is a circular process that can be seen as a mini-max game between the generator and the hybrid discriminator.

### 3.2. The generator for medical image report generation

Fig. 4 shows the generator's structure. The medical report generation procedure controlled by the generator can be described as follows.

**The visual feature of the given medical images.** Given a set of medical images$I = \{i_1, ..., i_l\}$belonging to a patient, a pre-trained CNN model like ResNet-50 takes it as the input to extract visual features expressed as feature matrix $X \in R^D$.

$$X = Concat(f_{CNN}(I)) \tag{1}$$

**The process of "topics to sentences".** A complete medical image report consists of multiple sentences, and each sentence focuses on one specific topic. As a result, we first obtain a number of topics according to the visual features given above. As shown in Fig. 4, the sentence LSTM included in the hierarchical LSTM is a single-layer LSTM that takes visual features as its input and then outputs the topic vectors $p \in R^K$ that serve as the input of another part of the hierarchical LSTM named word LSTM. Specifically, the visual features are first input into the LSTM to generate the hidden $h_{sent}^{(s)}$ state in the current time step $s$, and then the linear layer is used to combine the hidden state and the visual features to generate the topic vectors $p^{(s)}$. The procedure can be expressed as follows:

$$h_{sent}^{(s)} = LSTM(X), \tag{2}$$

$$p^{(s)} = tanh\left(W_{p,h_{sent}} h_{sent}^{(s)} + W_{p,X} X\right), \tag{3}$$

where $W_{p,h_{sent}}$ and $W_{p,X}$ are weight parameters generated through batch normalization. Moreover, a linear layer is used to control the number of the generated topics, which can be described as follows:

$$p\left(STOP|h_{sent}^{(s-1)}, h_{sent}^{(s)}\right) = \propto exp W_{stop} tanh\left(W_{stop,s-1} h_{(sent)}^{(s-1)} + W_{stop,s} h_{(sent)}^{(s)}\right), \tag{4}$$

where $p(\cdot)$ is a probability about "whether stop", $h_{(sent)}^{(s-1)}$and $h_{(sent)}^{(s)}$ are the previous state and the current state, respectively, and $W_{stop}$, $W_{stop,s-1}$ and $W_{stop,s}$ are the parameter matrices. It is to be noted that when $p(\cdot)$ is greater than a predefined threshold, the sentence LSTM will stop generating topic vectors and the whole hierarchical LSTM will stop decoding. After a certain number of topic vectors $p \in R^K$ are generated by the sentence LSTM, the word LSTM takes them and the special token "START" as the first input [38], and then outputs several sequences of words that from a complete medical image report. In this part, the hidden state $h_{word} \in R^W$ is directly regarded as the distribution over all words in the dictionary:

$$s(word|h_{word}) \propto exp(W_{out} h_{word}), \tag{5}$$

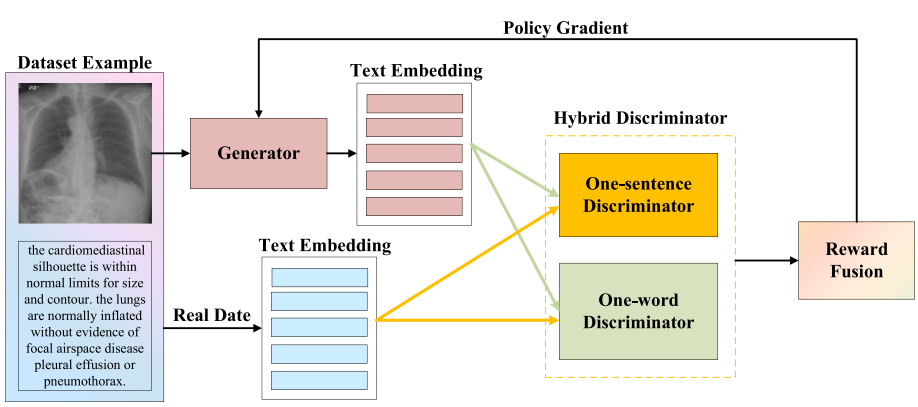



**Fig. 3.** The whole structure of the HDGAN model. It contains two main parts, namely, generator and hybrid discriminator. Hybrid discriminator contains two different discriminators, namely, one-sentence discriminator and one-word discriminator. The "Reward Fusion" including the outputs of the one-sentence discriminator and one-word discriminator can be regarded as an "update guide" for generator via policy gradient. The detailed structure of the generator, one-sentence discriminator and one-word discriminator will be introduced in the following chapters.

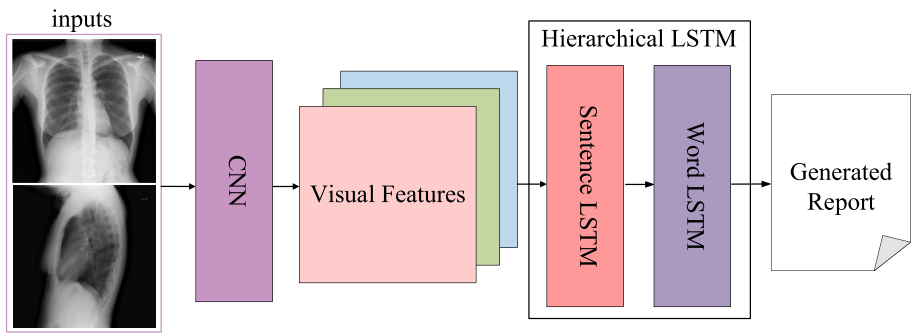



**Fig. 4.** The generator of the HDGAN model.

where $s(\cdot)$ is the distribution over all words, and $W_{out}$ is the weight matrix. The number of sentences is equal to the number of topics. Finally, all generated sentences are simply concatenated to produce a complete report.

### 3.3. Hybrid discriminator for medical image report generation

Deep neural network (DNN) [39], convolutional neural network (CNN) [40], recurrent convolutional neural network (RCNN) [41] have been extensively applied in handling complex text classification tasks. In this section, the hybrid discriminator that consists of a one-sentence discriminator and a one-word discriminator will be introduced.

As shown in Fig. 5, the one-sentence discriminator is a combined structure including an LSTM network and a linear layer for completing binary classification tasks. The output of the one-sentence discriminator is the predicted probability of a complete sentence being real or fake. The detailed process is described below. First, we represent the input sequence of $x_1, x_2, x_3, \ldots, x_n$ as:

$$E_{1:N} = e_1 \oplus e_2 \oplus \ldots \oplus e_n, \tag{6}$$

where $e_n$ is a k-dimensional embedding, and $\oplus$ is the symbol of the concatenation operator used to build the matrix $E_{1:N} \in R^{n\times k}$. Then, the embedding matrix $E$ is input into the one-sentence discriminator.

$$out = FC(LSTM(E)), \tag{7}$$

where *out* is a binary class output, and *FC* is a fully connected layer.

As shown in Fig. 6, the core of the one-word discriminator is composed of multiple parallel convolutional layers, and each convolutional layer is responsible for identifying the eigenvalues of a specified word. Most of the previous works tackle the execution of classification tasks for the entire sequence rather than a single word. In our work, we also pay attention to the result of the discriminator to determine whether a single word is real or fake. The detailed process is described below. First, the embedding $E$ is obtained by Eq. (6), then the filter $w \in R^{l\times k}$ first performs the convolutional operation to handle $l$ words in its receptive field and then outputs a new feature map:

$$m_i = \lambda(w \otimes E_{i:i+l-1} + b), \tag{8}$$

where $\lambda$ is a non-linear function, $b$is a bias term, and $\otimes$ operator is the summation of element-wise production. In this paper, different numbers of filters and different sizes of filters are used to find the optimal one-word discriminator structure. Secondly, a certain number of features are input into the max-pooling layer, which can be expressed as $\widetilde{m}_i = maxm_1, \ldots, m_{N-l+1}$. Finally, a fully connected layer with ReLU activation is applied over the output of the max-pooling layer to output the probability of the input sequence will being real or fake. During the training, the optimization target is to minimize the cross entropy between the ground truth label and the predicted probability. The equation can be formulated as follows:

$$\min_{\Phi} -E_{Y\,p_{data}}\left[logD_{\phi+\phi_1}(Y)\right] - E_{Y\,G_\theta}\left[log\left(1 - D_{\phi+\phi_1}(Y)\right)\right], \tag{9}$$

where $D_{\phi+\phi_1}$ denotes the representations of the two types of discriminators, $Y$is the input sequence, $Y\,p_{data}$denotes the sequence from the dataset named real data, and $Y\,G_\theta$ denotes the sequence generated from the generator named fake data.

### 3.4. HDGAN via policy gradient

In many studies, GANs have been used to generate a sequence since they were proposed. For example, SeqGAN [33] proposed a method that used the RL algorithm and considered the estimated probability of a sentence being real by the discriminator as the reward. The reward can be expressed by the equation below.

$$Q_{D_\phi}^{G_\theta}(a = y_T, s = Y_{1:T-1}) = D_\phi(Y_{1:T}), \tag{10}$$

In this paper, we take every sentence as the input of the one-sentence discriminator and every single word as the input of the one-word discriminator and then consider the estimated probability $D_{\phi_1}(Y_{t:t+1}^n)$, $D_\phi(Y_{1:t-1}^n)$ as the reward. Formally, we have

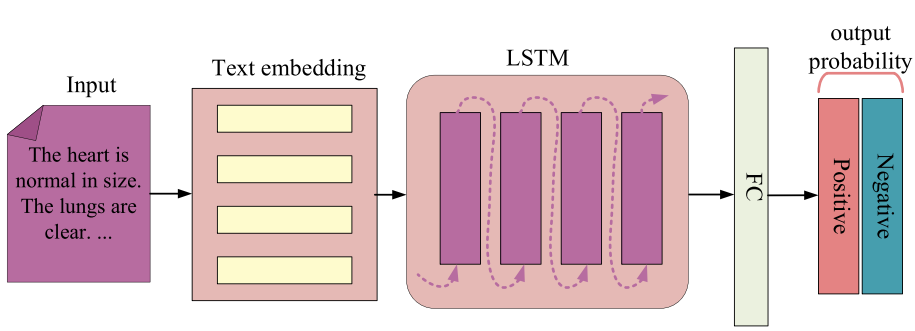



**Fig. 5.** The one-sentence discriminator of the HDGAN model.

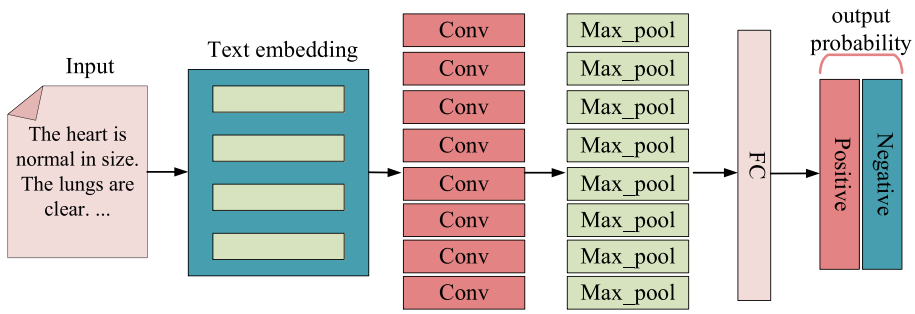



**Fig. 6.** The one-word discriminator of the HDGAN model.

$$Q^{G_\theta}_{D_\phi+D_{\phi_1}}(a=y_t, s=Y_{1:t-1}) = \sum_{t=1}^{T} D_\phi(Y_{1:t-1}) + D_{\phi_1}(Y_{t:t+1}), \tag{11}$$

where $D_\phi$ and $D_{\phi_1}$ are the one-sentence discriminator and one-word discriminator, respectively. We take the sum of $D_{\phi_1}(Y^n_{t:t+1})$ and $D_\phi(Y_{1:t-1})$ as the reward because it can be dynamically updated to further improve the generator iteratively. Once a set of more realistic medical image report sequences are generated, the hybrid discriminator should be retrained according to Eq. (8). After obtaining the new hybrid discriminator, we get ready to update the generator. The adopted policy-based method relies upon optimizing a parametrized policy to directly maximize the one-word and one-sentence rewards. The gradient of the objective function $J(\theta)$ about the generator's parameters $\theta$ can be derived as follows:

$$\nabla_\theta J(\theta) = \sum_{t=1}^{T} E_{Y_{1:t-1}\ G_\theta}\left[\sum_{y_t\in y} \nabla_\theta G_\theta(y_t|Y_{1:t-1}) \cdot Q^{G_\theta}_{D_\phi+D_{\phi_1}}(y_t, Y_{1:t-1}),\right. \tag{12}$$

where $Y_{1:t-1}$ is the observed intermediate state sampled from$G_\theta$. $E[\cdot]$, which can be approximated by the sampling method. Then, the generator's parameter can be updated as follows:

$$\theta \leftarrow \theta + lr_s \nabla_\theta J(\theta), \tag{13}$$

where $lr_s$ is the learning rate at the s-step.

**Algorithm 1.** HDGAN model training.

**Input:** input parameters generator $G_\theta$, discriminators $D_\phi$ and $D_{\phi_1}$, dataset $D_s$ includes medical image $I$ and reports sequence $S = X_{1:N}$
**Output:** medical image reports

```
Initialize weights of G_θ , D_φ and D_φ1 randomly
Pre-train G_θ on D_s using MLE
Generating negative samples by G_θ for training D_φ
Pre-train D_φ, D_φ1 by minimizing the cross entropy loss
repeat
for g-steps do
    Generate a report sequence Y_{1:T} = (y_1, ..., y_T) G_θ
    Compute Q^{G_θ}_{D_φ+D_φ1} by Eq 17.
    Update G_θ by Eq 19.
for t-steps do
    Use current G_θ to generate negative samples and combine with given positive examples D_s
    Train one-sentence discriminator D_φ for d-steps by Eq 15.
    Train one-word discriminator D_φ1 for d-steps by Eq 15.
until HDGAN converges
return result
```

### 3.5. Model learning

Algorithm 1 shows the detailed training process. At the beginning of training, the MLE (maximum likelihood estimation) is used to pre-train the generator using the training set, and then the supervised method is used to pre-train the hybrid discriminator. After pre-training, the generator and hybrid discriminator are trained alternately. After the generator has been trained for g-steps, the hybrid discriminator needs to be retrained periodically to keep pace with the generator. When training the hybrid discriminator, positive examples are taken from the given dataset and negative examples are generated by the generator. To keep the balance, the number of generated negative examples is the same as the positive examples in every epoch. To reduce the variability of the estimation, we use different sets of negative examples combined with positive examples.

## 4. Experiment

### 4.1. Dataset

Experiments are conducted on the medical image dataset named IU-X-ray, MIMIC-CXR and COV-CTR to evaluate our proposed model. The specific division is shown in Table 1.

**IU-X-ray** [42] The full name of this dataset is Indiana University Chest X-ray Collection, which consists of 7470 pairs of images and reports. Each report consists of impressions, findings, and tags.

**MIMIC-CXR** [43] The MIMIC-CXR is a large-scale dataset containing 377,110 chest X-ray images and 227,835 reports. The official split of MIMIC-CXR dataset is used in this paper, which includes 368,960 in the training set, 2991 in the validation set, and 5159 in the test set.

**COV-CTR** [44] The COV-CTR is a recently published CT COVID-19 dataset containing 728 images (349 for COVID-19 and 379 for Non-COVID) collected from published papers and their corresponding paired reports. It is important to note that this dataset does not provide multiple disease labels beyond COVID-19 and Non-COVID categories.

In this paper, the following methods are used to process the IU-X-ray, MIMIC-CXR and COV-CTR dataset: (1) Convert all tokens to lowercases; (2) Remove all of the non-alpha tokens, including punctuations; and (3) Delete excessively long/short sentences and keep the sentence length within the range of 4 to 8.

### 4.2. Implementation details

The model's dimensions of all hidden states and word embedding are

**Table 1**
The introduction of IU-X-ray, MIMIC-CXR and COV-CTR datasets.

| Datasets | Component | Train set | Validation set | Test set |
|---|---|---|---|---|
| IU-X-ray | Patient | 2770 | 395 | 790 |
| | Image | 5226 | 748 | 1496 |
| | Report | 2770 | 395 | 790 |
| MIMIC-CXR | Patient | 64,586 | 500 | 293 |
| | Image | 368,960 | 2991 | 5159 |
| | Report | 222,758 | 3269 | 3269 |
| COV-CTR | Image | 508 | 110 | 108 |
| | Report | 508 | 110 | 108 |

512. We pre-train the generator, one-sentence discriminator and one-word discriminator for 100, 50 and 50 epochs.

(1) **Generator.** First, the ResNet50, from which the last two layers are removed, is used to extract the visual features. Second, we train the model by using cross-entropy loss and using the Adam optimizer for parameter learning. Finally, the learning rates of the CNN and hierarchical LSTM are 1e-5 and 5e-4 respectively. The weights ($\lambda_{tag}$ $\lambda_{stop}$and $\lambda_{word}$)of different losses are set to 100, 10, and 1 respectively.
(2) **One-sentence Discriminator.** LSTM and fully connected network are used to perform the text classification task. At the training stage, the number of LSTM layers is set to 2.
(3) **One-word Discriminator.** 8 parallel convolution layers are used in the one-word discriminator. In each layer, we set the channel and kernel sizes to different values. For example, the channel number is set to 100, 200, 200, 200, 100, 200, 100, and 200, and the kernel size is set to 1–8.

### 4.3. Baselines

We compare our proposed model with several state-of-the-art image captioning models, such as S-ATT [2], CNN-RNN [45], LRCN [46], KERP [6] and ReMRG-XR [29]. To verify the efficiency of our proposed model, we also compare our model with the CoAtt [47] model. These baseline methods and results are reported in Tables 2–4

- **CNN-RNN** [45] The classic encoder-decoder architecture used for image captioning tasks. The CNN is the encoder, and the RNN is the decoder.
- **S-ATT** [2] It is an image captioning model with semantic attention. The top-down and bottom-up methods are combined. Attention is selectively paid to semantic concept proposals, which are then merged into the output of the hidden layer and the RNN. The selection and integration mentioned above form a feedback linking top-down and bottom-up calculations.
- **LRCN** [46] The Long-term Recurrent Convolutional Network (LRCN) is an end-to-end trainable structure for image recognition and captioning, which combines a CNN with an LSTM. It can process variable-length input sequences and support variable-length sentence output, including the generation of full-length descriptive sentences.
- **KERP** [6] The KERP model reconciles traditional knowledge and retrieval-based methods with modern learning-based methods for accurate and robust medical report generation. The core of the KERP is a proposed generic implementation unit called the Graph Transformer (GTR). The GTR dynamically transforms high-level semantics between graph-structured data in multiple domains, such as knowledge graphs, images, and sequences.
- **ReMRG-XR** [29] The ReMRG-XR model includes an x-linear attention module that is used to explore high-order feature interactions and achieve multi-modal reasoning, while the repetition penalty mechanism is used to apply penalties to repeated terms during the model's training process.
- **CoAtt** [47] This is a multi-task learning model for medical image report generation task that predicts tags and generates long paragraphs. The model uses the co-attention mechanism to determine the locations of lesions and uses a hierarchical LSTM to generate long paragraphs.
- **R2Gen** [27] The R2Gen uses a relational memory to record key information of the generation process, applys a memory-driven conditional layer normalization to incorporate the memory into the decoder of Transformer.
- **CMCL** [53] In the CMCL, the curriculum learning is integrated into the process of medical report generation, which enables the models to gradually proceed from easy samples to more complex ones in training and uses the limited medical data to mitigate data bias.
- **KM** [54] The KM includes a knowledge-enhanced multi-head attention mechanism, and integrated general knowledge and specific knowledge into the Transformer.

### 4.4. Quantitative results

We evaluate the paragraph generation results using the standard image captioning evaluation tool that includes the following metrics: BLEU [48],

METEOR [49], ROUGE [50], and CIDEr [51]. Generally, those scores are higher if the model's output is correct and sentence structure aligns better with the real medical image report.

As shown in Table 2, the HDGAN model performs better and brings more significant improvements in all evaluation metrics than CNN-RNN, S-ATT, LRCN, R2Gen and CMCL. Moreover, in CIDEr, the HDGAN performs better than CoAtt.

As shown in Table 3, we compare the performance of the HDGAN with other models on the MIMIC-CXR dataset. The experimental results show that, compared with previous work, the HDGAN surpasses them in the metrics of ROUGE and is slightly weaker than the recently published model in terms of BLEU. However, R2Gen [27] adopted the Transformer as the Decoder to generate medical reports and we adopt the LSTM as the Decoder. As a full-attention model with excellent performance in the field of natural language processing, the Transformer has a large number of parameters. By comparison, our proposed model is more lightweight and has lower requirements for hardware devices.

In addition, we compare the performance of the HDGAN with other models on the COV-CTR datasets (shown in Table 4). Since the methods we compared did not use this new dataset, their experimental results are lacking. Therefore, we can only reproduce the R2Gen method, which provides the code and does not need labels. But the high scores for various metrics still verify the success of our method in COV-CTR datasets.

### 4.5. Ablation studies

In order to demonstrate contributions from different components in

**Table 2**
Comparison with some medical report generation models on the IU-X-ray.

| Dataset | Model | B-1 | B-2 | B-3 | B-4 | METEOR | ROUGE | CIDEr |
|---|---|---|---|---|---|---|---|---|
| IU-X-ray | CNN-RNN [45] | 0.237 | 0.156 | 0.092 | 0.075 | 0.132 | 0.178 | 0.171 |
| | S-ATT [2] | 0.251 | 0.158 | 0.089 | 0.066 | 0.136 | 0.193 | 0.155 |
| | LRCN [46] | 0.269 | 0.159 | 0.092 | 0.083 | 0.155 | 0.189 | 0.190 |
| | KERP [6] | 0.482 | 0.325 | 0.226 | 0.162 | – | 0.339 | 0.280 |
| | ReMRG-XR [29] | 0.402 | 0.276 | 0.196 | 0.140 | 0.184 | 0.357 | 0.433 |
| | CoAtt* [47] | 0.517 | 0.386 | 0.306 | 0.247 | 0.217 | 0.447 | 0.327 |
| | R2Gen [27] | 0.470 | 0.304 | 0.219 | 0.165 | 0.187 | 0.371 | – |
| | CMCL [53] | 0.473 | 0.305 | 0.217 | 0.162 | 0.186 | 0.378 | – |
| | KM [54] | 0.496 | 0.327 | 0.238 | 0.178 | – | 0.381 | 0.382 |
| | **HDGAN** | 0.477 | 0.318 | 0.226 | 0.167 | 0.190 | 0.417 | 0.426 |

**Table 3**
Comparison with some medical report generation models on the MIMIC-CXR.

| Dataset | Model | B-1 | B-2 | B-3 | B-4 | METEOR | ROUGE |
|---|---|---|---|---|---|---|---|
| MIMIC-CXR | ST [1] | 0.299 | 0.184 | 0.121 | 0.084 | 0.124 | 0.263 |
| | Ada-Att [52] | 0.299 | 0.185 | 0.124 | 0.088 | 0.118 | 0.266 |
| | R2Gen [27] | 0.353 | 0.218 | 0.145 | 0.103 | 0.142 | 0.277 |
| | CMCL [53] | 0.344 | 0.217 | 0.140 | 0.097 | 0.133 | 0.281 |
| | KM [55] | 0.363 | 0.228 | 0.156 | 0.115 | – | 0.284 |
| | **HDGAN** | 0.352 | 0.195 | 0.122 | 0.084 | 0.115 | 0.290 |

**Table 4**
Comparison with some medical report generation models on the COV-CTR.

| Dataset | Model | B-1 | B-2 | B-3 | B-4 | METEOR | ROUGE |
|---|---|---|---|---|---|---|---|
| COV-CTR | R2Gen [27] | 0.660 | 0.568 | 0.495 | 0.436 | 0.344 | 0.602 |
| | **HDGAN** | 0.765 | 0.705 | 0.656 | 0.617 | 0.442 | 0.758 |

our proposed HDGAN model, we conduct an extensive ablation study. The results are given in Table 5. As mentioned above, we use the one-sentence reward and one-word reward in the model, respectively, to verify the effectiveness of the two types of rewards. The model with the one-word reward performs better than the model with the one-sentence reward. The reason is that the former provides more word diversity information, while the latter provides sentence structure information. Compared to the sentence structure information, the results of the medical image report generation task pay more attention to the correctness of words. Furthermore, we integrate the one-sentence reward and the one-word reward into the HDGAN model at the same time to achieve a better performance. There are two advantages in doing so: (1) The ability of the HDGAN model to learn sentence structure information of medical image reports is improved through the mechanism of one-sentence reward. It enables the model to generate more normative medical image reports. (2) The ability of the HDGAN model to learn diverse professional vocabulary is improved through the mechanism of one-word reward. Subsequently, accurate professional descriptions are generated and included in medical image reports. Therefore, the HDGAN model not only pays attention to the word diversity information, but it also captures the sentence structure information. The experimental results demonstrate that the combination of one-sentence reward and one-word reward is effective.

### 4.6. Human evaluation

We randomly select 50 images from the dataset and invite three radiologists to evaluate the corresponding outputs of our HDGAN model and CoAtt. They are encouraged to select a more accurate result from each pair in terms of content coverage, the accuracy of specific terminology, and language fluency. The human evaluation results are presented in Table 6. These results demonstrate that our method is capable of generating accurate and semantic-coherent reports.

### 4.7. Qualitative comparison

An illustration of medical image reports generated by two models (CoAtt model, HDGAN model) on IU-X-ray dataset is shown in Fig. 7. It can be seen that different sentences in the medical image reports correspond to different topics. For example, the first sentence of the report generated from the first image is a high-level description of the medical image, while other sentences are associated with the specific areas of the image (e.g. “lung”, “heart”) respectively. The CoAtt model detects few abnormalities and even wrong abnormalities and generates reports with complex sentences. In contrast, our HDGAN model can correctly describe many true abnormalities and generate reports with brief sentences. This comparison demonstrates that our HDGAN model is better at capturing abnormalities.

**Table 6**
Human evaluation (Hit).

| Models | Co-Att | HDGAN |
|---|---|---|
| Hit(%) | 19 | 48 |

As can be seen from the first image reports’ green parts and the second image reports’ blue parts, the descriptions generated by the HDGAN model are brief and correct, while the CoAtt model tends to generate more complicated and repeated sentences. As can be seen from the green parts in the third image reports, the HDGAN model finds the abnormality of “left-sided pleural effusion”, while the report generated by CoAtt model does not show the abnormality correctly. As can be seen from the green parts in the fourth image reports, the HDGAN model also successfully detects the area (“lungs”) that is abnormal (“expanded”), while the CoAtt model fails to find enough correct abnormalities correctly. In general, the colored parts of the Fig. 7 shows the correct results generated by the models. Obviously, our HDGAN model generates more accurate descriptions than the CoAtt model.

### 4.8. The influence of the combination of two types of reward

The one-sentence discriminator and one-word discriminator transfer different reward signals to the generator. The reward from the one-sentence discriminator represents the learned sentence structure information, and the reward from the one-word discriminator represents the learned word diversity information. Therefore, the different combinations of these two types of rewards can transfer different “signals” to update the generator and have different effects on the generator’s performance. As shown in Fig. 8, we use the IU-X-ray dataset to conduct

**Table 5**
Ablation study to demonstrate contributions from different proposed components on IU-X-ray dataset. “s-reward” means that we only use the reward from one-sentence discriminator to update generator; “w-reward” means that we only use the reward from one-word discriminator to update generator.

| Model | BLEU-1 | BLEU-2 | BLEU-3 | BLEU-4 | METEOR | ROUGE | CIDEr |
|---|---|---|---|---|---|---|---|
| s-reward | 0.455 | 0.299 | 0.214 | 0.160 | 0.181 | 0.392 | 0.410 |
| w-reward | 0.454 | 0.297 | 0.212 | 0.157 | 0.178 | 0.390 | 0.376 |
| HDGAN | **0.477** | **0.318** | **0.226** | **0.167** | **0.190** | **0.417** | **0.426** |

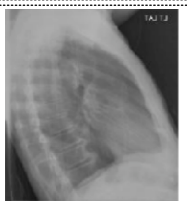
| Input Images | Groud Truth | CoAtt | HDGAN |
|---|---|---|---|
| | no acute pulmonary abnormality. the lungs and pleural spaces show no acute abnormality. heart size and pulmonary vascularity within normal limits. | no acute cardiopulmonary disease. the heart pulmonary xxxx and mediastinum are within normal limits. the heart is normal in size. **the lungs are free of focal airspace disease.** no pleural effusion or pneumothorax is seen. no pleural effusion. | no acute cardiopulmonary abnormality. the heart is normal in size. **the lungs are clear.** no pneumothorax or pleural effusion. no pleural effusion or pneumothorax is seen. no acute bony. |
| | no acute cardiopulmonary disease. the lungs appear clear. the heart and pulmonary xxxx appear normal. the pleural spaces are clear. mediastinal contours are <unk> | no acute cardiopulmonary abnormality. the heart is normal in size. the lungs are clear. **no pneumothorax or pleural effusion. no pleural effusion or pneumothorax is seen. no pleural effusion or pneumothorax.** | **no acute cardiopulmonary disease** . the heart is normal in size. **the lungs are clear.** no focal consolidation pneumothorax or pleural effusion. **no pneumothorax or pleural effusion.** no acute bony . |
| | ap and lateral views were obtained. Bibasilar atelectasis and small **leftsided pleural effusion.** stable cardiomegaly. no pneumothorax. mild pulmonary vascular congestion. | c a r diomegaly with elevated right hemidiaphragms and no acute findings. no acute radiographic cardiopulmonary process. there is a moderate sized right pleural effusion. | the heart is normal in size. the lungs are clear. no focal airspace consolidation pneumothorax. no pneumothorax or pleural effusion. **there is a pleural effusion within the left side.** |
|  | **no active disease.** the heart and lungs have xxxx xxxx in the interval. **both lungs are clear and expanded.** heart and mediastinum normal. | no acute cardiopulmonary findings. the heart is normal in size. the mediastinum is unremarkable. no pneumothorax or pleural effusion. no acute osseous <unk> | **no active disease.** the heart is normal in size. **lungs are clear and expanded.** no pneumothorax or pleural effusion is seen. |

**Fig. 7.** Illustration of the generated reports.

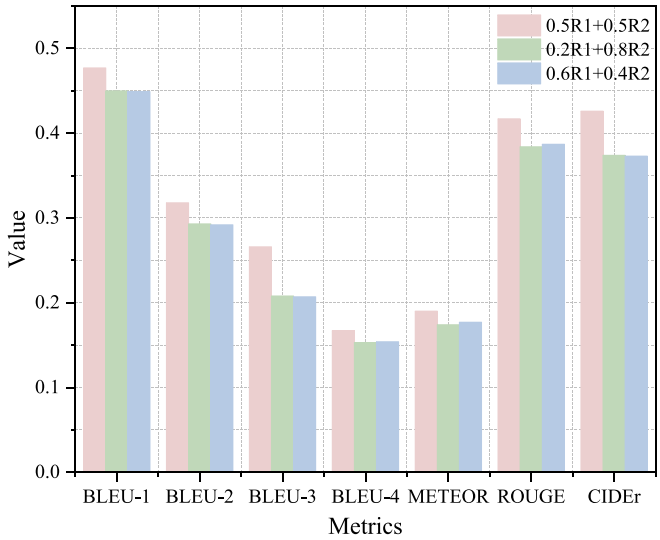



**Fig. 8.** The different combinations of two rewards. R1 represents the one-sentence discriminator's reward signal, R2 represents the one-word discriminator's reward signal.

experiments on different weight combinations of the two types of rewards. After many experiments, we set the weights of the two types of rewards to 0.5 and 0.5. With this setting, we give equal attention to the one-sentence reward and one-word reward, and the generator achieves the best performance.

## 5. Conclusion

In this paper, we proposed the HDGAN model for automatic generation of medical image reports. The main works include the following three aspects: (1) Integrating the GAN into the model to improve the evaluation indexes and generate more realistic reports, (2) modeling the data generator as a stochastic policy in RL and altering the RL reward signal from the GAN discriminator judged on both a complete sentence and single word, and (3) using the hybrid discriminator to transfer two types of reward signals (one-sentence reward and one-word reward) to the generator. The one-sentence reward provides sentence structure information for the generator, while the one-word reward provides word diversity information for the generator. These contributions achieved more attention to text processing, addressed the difficulty of gradient updating from the discriminator to the generator, and learned more effectively to report structural information. The rationality and validity of the proposed model have been verified through several experiments.

There are still some limitations in processing discrete sequences with the HDGAN model. Specifically, the discriminator generates rewards based on a complete sequence. Therefore, the generation of the first half of the sequence will be penalized if the second half of the generated sequence is wrong. The model needs to be further improved and optimized in the future.

## CRediT authorship contribution statement

**Junsan Zhang:** Conceptualization, Methodology, Project administration, Supervision, Writing – review & editing, Investigation, Funding acquisition. **Ming Cheng:** Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing. **Qiaoqiao Cheng:** Data curation, Methodology, Writing – original draft. **Xiuxuan Shen:**

Writing – review & editing. **Yao Wan:** Supervision, Writing – review & editing. **Jie Zhu:** Formal analysis, Funding acquisition. **Mengxuan Liu:** Formal analysis.

## Declaration of competing interest

We declare that we do not have any commercial or associative interest that represents a conflict of interest in connection with the work submitted.

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## References

[1] Vinyals, Toshev A, Bengio S, Erhan D. Show and tell: A neural image caption generator. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2015. p. 3156–64.
[2] You Q, Jin H, Wang Z, Fang C, Luo J. Image captioning with semantic attention. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2016. p. 4651–9.
[3] Monshi M, Poon J, Chung V. Deep learning in generating radiology reports: a survey. Artif Intell Med 2020;106:101878.
[4] Delrue L, Gosselin R, Ilsen B, Van Landeghem A, de Mey J, Duyck P. Difficulties in the interpretation of chest radiography. In: Comparative interpretation of CT and standard radiography of the chest. Springer; 2011. p. 27–49.
[5] Li CY, Liang X, Hu Z, Xing EP. Hybrid retrieval-generation reinforced agent for medical image report generation. arXiv preprint, arXiv:1805.08298; 2018.
[6] Li CY, Liang X, Hu Z, Xing EP. Knowledge-driven encode, retrieve, paraphrase for medical image report generation. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 33; 2019. p. 6666–73.
[7] Goodfellow IJ, Pouget-Abadie J, Mirza M, Xu B, Warde-Farley D, Ozair S, et al. Generative adversarial networks. arXiv preprint, arXiv:1406.2661; 2014.
[8] Sutton RS, Barto AG. Reinforcement learning: An introduction. MIT press; 2018.
[9] Huang H, Yu PS, Wang C. An introduction to image synthesis with generative adversarial nets. arXiv preprint, arXiv:1803.04469; 2018.
[10] Li C, Su Y, Liu W. Text-to-text generative adversarial networks. In: In: 2018 international joint conference on neural networks (IJCNN). IEEE; 2018. p. 1–7.
[11] Yang Z, Yuan Y, Wu Y, Salakhutdinov R, Cohen WW. Review networks for caption generation. arXiv preprint, arXiv:1605.07912; 2016.
[12] Xu K, Ba J, Kiros R, Cho K, Courville A, Salakhudinov R, et al. Show, attend and tell: neural image caption generation with visual attention. In: International conference on machine learning. PMLR; 2015. p. 2048–57.
[13] Rennie SJ, Marcheret E, Mroueh Y, Ross J, Goel V. Self-critical sequence training for image captioning. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2017. p. 7008–24.
[14] Anderson P, He X, Buehler C, Teney D, Johnson M, Gould S, et al. Bottom-up and top-down attention for image captioning and visual question answering. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2018. p. 6077–86.
[15] Dognin P, Melnyk I, Mroueh Y, Ross J, Sercu T. Adversarial semantic alignment for improved image captions. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2019. p. 10463–71.
[16] Guo L, Liu J, Yao P, Li J, Lu H. Mscap: multi-style image captioning with unpaired stylized text. In: Proceedings of the IEEE/CVF s; 2019. p. 4204–13.
[17] Chen S, Jin Q, Wang P, Wu Q. Say as you wish: Fine-grained control of image caption generation with abstract scene graphs. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition; 2020. p. 9962–71.
[18] Kim D-J, Choi J, Kweon IS, et al. Dense relational image captioning via multi-task triple-stream networks. arXiv preprint, arXiv:2010.03855; 2020.
[19] Zhang W, Nie W, Li X, Yu Y. Image caption generation with adaptive transformer. In: 2019 34rd Youth Academic Annual Conference of Chinese Association of Automation (YAC); 2019.
[20] Vijayakumar A, Cogswell M, Selvaraju R, Sun Q, Lee S, Crandall D, et al. Diverse beam search for improved description of complex scenes. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 32; 2018.
[21] Shetty R, Rohrbach M, Anne Hendricks L, Fritz M, Schiele B, et al. Proceedings of the IEEE international conference on computer vision. 2017. p. 4135–44.
[22] Wang Z, Wu F, Lu W, Xiao J, Li X, Zhang Z, et al. Diverse image captioning via grouptalk. in: IJCAI; 2016. p. 2957–64.
[23] Aneja J, Agrawal H, Batra D, Schwing A. Sequential latent spaces for modeling the intention during diverse image captioning, arXiv. 2019.
[24] Mahajan S, Gurevych I. Latent normalizing flows for many-to-many cross-domain mappings. In: International Conference on Learning Representations; 2019.
[25] Wang X, Peng Y, Lu L, Lu Z, Summers RM. Tienet: Text-image embedding network for common thorax disease classification and reporting in chest x-rays. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2018. p. 9049–58.
[26] Xue Y, Xu T, Long LR, Xue Z, Antani S, Thoma GR, et al. Multimodal recurrent model with attention for automated radiology report generation. In: International conference on medical image computing and computer-assisted intervention. Springer; 2018. p. 457–66.
[27] Chen Z, Song Y, Chang TH, Wan X. Generating radiology reports via memory-driven transformer. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP); 2020.
[28] Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need, arXiv. 2017.
[29] Xu W, Qi C. Reinforced medical report generation with x-linear attention and repetition penalty. 2020. arXiv:2011.07680.
[30] Liu F, Yin C, Wu X, Ge S, Zhang P, Sun X. Contrastive attention for automatic chest x-ray report generation. In: Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021; 2021. p. 269.
[31] Han Z, Wei B, Xi X, Chen B, Yin Y, Li S. Unifying neural learning and symbolic reasoning for spinal medical report generation. Med Image Anal 2021;67:101872. https://doi.org/10.1016/j.media.2020.101872. https://www.sciencedirect.com/science/article/pii/S136184152030236X.
[32] Sutton RS, McAllester DA, Singh SP, Mansour Y, et al. Policy gradient methods for reinforcement learning with function approximation. In: NIPs. Vol. 99. Citeseer; 1999. p. 1057–63.
[33] Yu L, Zhang W, Wang J, Yu Y. Seqgan: Sequence generative adversarial nets with policy gradient. In: Proceedings of the AAAI conference on artificial intelligence. Vol. 31; 2017.
[34] Zhang Y, Gan Z, Carin L. Generating text via adversarial training. In: NIPS workshop on Adversarial Training. Vol. 21; 2016.
[35] Na L, Yang G, An D, Xin QXGu. Mode collapse and regularity of optimal transportation maps. 2019. arXiv:1902.02934.
[36] Zhang Y, Gan Z, Fan K, Chen Z, Henao R, Shen D, et al. Adver-sarial feature matching for text generation. In: International Conference on Machine Learning. PMLR; 2017. p. 4006–15.
[37] Fedus W, Goodfellow I, Dai AM. MaskGAN: Better text generation via filling in the _. In: Proc. Int. Confe. Learn. Representations; 2018. p. 1–16.
[38] Krause J, Johnson J, Krishna R, Fei-Fei L. A hierarchical approach for generating descriptive image paragraphs. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2017. p. 317–25.
[39] Vesely K, Ghoshal A, Burget L, Povey D. Sequence-discriminative training of deep neural networks. In: Interspeech. 2013; 2013. p. 2345–9.
[40] Rakhlin A. Convolutional neural networks for sentence classification. In: MS thesis. University of Waterloo; 2015.
[41] Lai S, Xu L, Liu K, Zhao J. Recurrent convolutional neural networks for text classification. In: Proceedings of the AAAI Conference on Artificial Intelligence. Vol. 29; 2015.
[42] Dina DF, Kohli MD, Rosenman MB, Shooshan SE, Laritza R, Sameer A, et al. Preparing a collection of radiology examinations for distribution and retrieval. Journal of the American Medical Informatics Association Jamia 2015;2:2.
[43] Johnson AE, Pollard TJ, Berkowitz SJ, Greenbaum NR, Lungren MP, Deng C-y, Mark RG, Horng S. Mimic-cxr, a de-identified publicly available database of chest radiographs with free-text reports. Sci Data 2019;6(1):1–8.
[44] Li M, Wang F, Chang X, Liang X. Auxiliary signal-guided knowledge encoder-decoder for medical report generation. 2019. arXiv preprint, arXiv:2006.03744.
[45] Xu K, Ba J, Kiros R, Cho K, Courville A, Salakhutdinov R, et al. Show, attend and tell: neural image caption generation with visual attention. Computer ence 2015: 2048–57.
[46] Donahue J, Hendricks LA, Rohrbach M, Venugopalan S, Guadarrama S, Saenko K, et al. Long-term recurrent convolutional networks for visual recognition and description, in. IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2015;2017:677–91.
[47] Jing B, Xie P, Xing E. On the automatic generation of medical imaging reports. arXiv preprint, arXiv:1711.08195; 2017.
[48] Papineni K, Roukos S, Ward T, Zhu W-J. Bleu: a method for automatic evaluation of machine translation. In: Proceedings of the 40th annual meeting of the association for computational linguistics; 2002. p. 311–8.
[49] Denkowski M, Lavie A. Meteor universal: Language specific translation evaluation for any target language. In: Proceedings of the ninth workshop on statistical machine translation; 2014.
[50] Lin C-Y. Rouge: a package for automatic evaluation of summaries. In: Text Summarization Branches Out; 2004. p. 74–81.
[51] Vedantam R, Lawrence Zitnick C, Parikh D. Cider: consensus-based image description evaluation. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2015. p. 4566–75.
[52] Lu J, Xiong C, Parikh D, Socher R. Knowing when to look: adaptive attention via a visual sentinel for image captioning. In: Proceedings of the IEEE conference on computer vision and pattern recognition; 2017. p. 375–83.
[53] Liu F, Ge S, Wu X. Competence-based Multimodal Curriculum Learning for Medical Report Generation[C]//ACL/IJCNLP 2021;1.
[54] Yang S, Wu X, Ge S, et al. Knowledge matters: chest radiology report generation with general and specific knowledge[J]. Med Image Anal 2022;102510.